

01. 能量守恒定律;law of conservation of energy

The energy can neither be created nor destroyed; rather, it can only be transformed or transferred from one form to another.

02. 动量守恒定律;law of conservation of momentum

The momentum of an isolated system remains constant. Momentum is therefore said to be conserved over time; that is, momentum is neither created nor destroyed, only transformed or transferred from one form to another.

03. 牛顿第一定律;Newton's First law of Motion

In an inertial frame of reference, an object either remains at rest or continues to move at a constant velocity, unless acted upon by a force.

04. 牛顿第二定律;Newton's Second law of Motion

In an inertial frame of reference, the vector sum of the forces F on an object is equal to the mass m of that object multiplied by the acceleration a of the object: $F = ma$.

05. 牛顿第三定律;Newton's Third law of Motion

When one body exerts a force on a second body, the second body simultaneously exerts a force equal in magnitude and opposite in direction on the first body.

06. 热力学第一定律;First law of thermodynamics

The total energy of an isolated system is constant; energy can be transformed from one form to another, but can be neither created nor destroyed.

07. 热力学第二定律;second law of thermodynamics

The second law of thermodynamics states that the total entropy of an isolated system can never decrease over time, and is constant if and only if all processes are reversible. Isolated systems spontaneously evolve towards thermodynamic equilibrium, the state with maximum entropy.

08. 热力学第三定律;third law of thermodynamics

The entropy of a system approaches a constant value as its temperature approaches absolute zero.

09. 热力学第零定律;Zeroth law of thermodynamics

The zeroth law of thermodynamics states that if two thermodynamic systems are each in thermal equilibrium with a third one, then they are in thermal equilibrium with each other.

(From Wikipedia)